

PRISMSPACE

Project Group 01

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Presentation roadmap

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Overview

While modern development tasks span code, browsers, databases, and **APIs** (Application Programming Interfaces), uncoordinated multi-agent workflows suffer from context fragmentation, runaway loops, and tool hallucinations. To address these challenges, PrismSpace introduces a unified developer workspace featuring **multi-agent orchestration**, sandboxed **MCP** (Model Context Protocol) tool execution with strict **ReAct** (Reasoning and Acting) loops, **HITL** (Human-in-the-Loop) safety, and **ORPO** (Odds Ratio Preference Optimization) preference alignment.

Objectives

Core Objectives

The primary objective is to enable structured task execution through **DAG** (Directed Acyclic Graph)-based task decomposition, **MCP** (Model Context Protocol)-enabled multi-agent **ReAct** (Reasoning and Acting) orchestration, and **ORPO** (Odds Ratio Preference Optimization) preference optimization. Additionally, the system ensures reliable and cost-effective operation by incorporating supervised **LLM** (Large Language Model) routing alongside Isolation Forest anomaly detection and human-in-the-loop validation.

Problem Statement

Problem Formulation

Because ungrounded **LLM** (Large Language Model) agents risk hallucinating tool results and executing unsafe actions, the system takes developer task prompts, workspace states, and **MCP** (Model Context Protocol)/**HITL** (Human-in-the-Loop) configurations as its primary **inputs**. Through an orchestration **method** integrating sandboxed **MCP ReAct** (Reasoning and Acting) loops, **ORPO** (Odds Ratio Preference Optimization), supervised **LLM** routing, and Isolation Forest anomaly detection with **HITL** oversight, it converts these tasks into dependable **outputs** consisting of verified artifacts, **ReAct** execution traces, and cost telemetry.

Multi-Objective Reward Generalization in Deep Reinforcement Learning

Citation	Summary	Advantages & Disadvantages
F. Cornalba, C. Disselkamp, D. Scassola, and C. Helf <i>Neural Computing and Applications</i> Year: 2024 Publisher: Springer Open Access	Investigates Multi-Objective Deep Reinforcement Learning (MODRL) for optimizing competing operational objectives (e.g., return vs. risk). Introduces reward generalization techniques that train pareto-optimal policies capable of dynamically adapting to shifting stakeholder preferences without re-training from scratch.	Advantages: ▶ Provides structured Pareto-optimal policy exploration across conflicting objectives. ▶ Generalizes across parameterized reward weights without full policy retraining. ▶ Directly informs multi-objective routing (latency vs. cost vs. quality) in PrismSpace. Disadvantages: ▶ High initial training sample complexity across multi-dimensional reward spaces. ▶ Evaluated primarily on continuous financial time-series data. ▶ Sensitive to hyperparameter tuning in policy entropy regularization.

Average Reward Adjusted Discounted Reinforcement Learning

Citation	Summary	Advantages & Disadvantages
M. Schneckenreither and G. Moser <i>Neural Computing and Applications</i> Year: 2025 Publisher: Springer Open Access	Proposes Average-Reward Adjusted Discounted RL (Reinforcement Learning) (ARAD-RL), bridging the gap between discounted and average reward criteria in infinite-horizon MDPs (Markov Decision Processes). Addresses systematic value estimation bias in standard discounted Q-learning by dynamically subtracting the estimated gain rate during Bellman updates.	Advantages: ▶ Eliminates infinite-horizon return explosion while preserving policy convergence. ▶ Improves long-term steady-state throughput in continuous scheduling tasks. ▶ Theoretically sound formulation for continuous agent lifecycle optimization. Disadvantages: ▶ Requires accurate empirical estimation of average reward baseline. ▶ Slow convergence in highly non-stationary transition environments. ▶ Adds variance to early-stage TD (Temporal Difference) error gradient estimates.

Deep Reinforcement Learning for Multi-Agent Coordination

Citation	Summary	Advantages & Disadvantages
K. Aina and S. Ha <i>Artificial Life and Robotics</i> Year: 2026 Publisher: Springer Open Access	Develops decentralized MARL (Multi-Agent Reinforcement Learning) algorithms for multi-agent coordination in constrained environments. Employs spatial-temporal observation encoding and localized interaction graphs to coordinate joint movements and resolve deadlocks without requiring centralized global state knowledge.	Advantages: ▶ Resolves shared-resource contention and deadlocks via decentralized policies. ▶ Spatial-temporal graph attention scales linearly with agent density. ▶ Applicable to concurrent tool-use contention across worker agents. Disadvantages: ▶ Tested in physical robotics grids rather than software API (Application Programming Interface) environments. ▶ Vulnerable to communication packet dropouts and sensor latency. ▶ Does not account for semantic language-based coordination.

Open-Ended Coordination Using Modular Open Policies

Citation	Summary	Advantages & Disadvantages
D. Rother, J. Pajarinen, J. Peters, and T. H. Weisswange <i>Autonomous Agents and Multi-Agent Systems</i> Year: 2025 Publisher: Springer Open Access	Presents a framework for open-ended ad hoc multi-agent coordination. Utilizes Modular Open Policies (MOP) where dynamic agent teams dynamically infer unseen teammate skill representations and synthesize blended coordination strategies without requiring pre-coordinated joint training.	Advantages: ▶ Enables zero-shot ad hoc coordination with unfamiliar teammate agent policies. ▶ Modular policy decomposition allows plug-and-play agent role assignment. ▶ Directly adopted in PrismSpace for dynamic worker swarm scaling. Disadvantages: ▶ Requires comprehensive pre-trained skill policy libraries for base roles. ▶ Evaluated primarily on benchmark games (Overcooked, LBF (Level-Based Foraging)). ▶ Action blending may introduce transient sub-optimal execution steps.

Improving Scalability of **MADRL** (Multi-Agent Deep Reinforcement Learning) with Suboptimal Human Knowledge

Citation	Summary	Advantages & Disadvantages
D. Liu, F. Ren, J. Yan, G. Su, W. Gu, and S. Kato <i>Autonomous Agents and Multi-Agent Systems</i> Year: 2026 Publisher: Springer Open Access	Addresses sample inefficiency in large-scale Multi-Agent DRL (Deep Reinforcement Learning) by integrating heuristic human guidance into exploration. Proposes an adaptive filtering mechanism that extracts actionable macro-actions from noisy or sub-optimal human demonstration traces, accelerating multi-agent policy learning.	Advantages: ▶ Accelerates policy exploration by up to 4x using noisy demonstration priors. ▶ Robust to suboptimal or conflicting human heuristic rules. ▶ Informs PrismSpace's bootstrap initialization from logged developer sessions. Disadvantages: ▶ Quality of learned policies depends on initial heuristic coverage. ▶ Requires complex threshold tuning to transition away from demonstrations. ▶ High memory footprint for maintaining dual replay buffers.

ϵ -Retraining Reinforcement Learning Algorithms

Citation	Summary	Advantages & Disadvantages
L. Marzari, C. Liu, P. L. Donti, and E. Marchesini <i>Autonomous Agents and Multi-Agent Systems</i> Year: 2026 Publisher: Springer Open Access	Investigates bounded policy fine-tuning through ϵ-retraining, preventing catastrophic forgetting when deep RL agents adapt to non-stationary environments. Establishes formal Lipschitz bounds on policy drift, ensuring safe continual learning in deployed agent pipelines.	Advantages: ▶ Provides mathematical guarantees against catastrophic policy divergence. ▶ Enables continuous telemetry-driven model fine-tuning in production. ▶ Crucial for stability in PrismSpace's ongoing user interaction learning. Disadvantages: ▶ Constrained optimization increases computation per gradient step. ▶ Conservative update bounds may slow adaptation to drastic distribution shifts. ▶ Theoretical guarantees rely on local smoothness assumptions.

Deep Multiagent Reinforcement Learning: Challenges and Directions

Citation	Summary	Advantages & Disadvantages
A. Wong, T. Bäck, A. V. Kononova, and A. Plaat <i>Artificial Intelligence Review</i> Year: 2023 Publisher: Springer Open Access	Surveys Deep Multiagent Reinforcement Learning (MARL), formalizing foundational challenges including non-stationarity, multi-agent credit assignment, and combinatorial action explosion. Evaluates Centralized Training with Decentralized Execution (CTDE) and communication architectures for modular orchestration.	Advantages: ▶ Comprehensive taxonomy of credit assignment and CTDE paradigms. ▶ Establishes formal theoretical criteria for communication protocols. ▶ Provides the theoretical foundation for PrismSpace's modular agent architecture. Disadvantages: ▶ Survey paper without empirical experimental evaluation. ▶ Focuses on discrete game simulations rather than API-driven software tasks. ▶ Does not cover LLM reasoning agents or human-in-the-loop approval gates.

HIAT (Human-in-the-Loop with Auxiliary Tasks): Human-in-the-Loop RL with Auxiliary Tasks

Citation	Summary	Advantages & Disadvantages
B. Niu, B. Luo, Y. Cui, X. Xu, Y. Zhao, and Y. Feng <i>Artificial Intelligence Review</i> Year: 2025 Publisher: Springer Open Access	Introduces HIAT (Human-in-the-Loop with Auxiliary Tasks), a framework leveraging human intervention signals as auxiliary self-supervised learning objectives. Models human corrections as feedback gradients that shape internal feature representations, minimizing human intervention frequency over time.	Advantages: ▶ Maximizes learning efficiency from scarce human intervention signals. ▶ Significantly reduces human cognitive load through auxiliary reward shaping. ▶ Informs PrismSpace's approval gate telemetry and learning from user edits. Disadvantages: ▶ Requires consistent human feedback to avoid biased auxiliary representations. ▶ Intervention timing latency can cause temporal mis-alignment in RL updates. ▶ Higher computational overhead during auxiliary task representation training.

ORPO (Odds Ratio Preference Optimization): Monolithic Preference Optimization without Reference Model

Citation	Summary	Advantages & Disadvantages
J. Hong, N. Lee, and J. Thorne <i>EMNLP</i> / <i>arXiv:2403.07691</i> Year: 2024 Area: LLM Preference Alignment	Core Idea (Single-Phase Alignment): Instead of multi-step RLHF, ORPO teaches desired answers while penalizing bad answers in one unified equation: $\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda \cdot \mathcal{L}_{\text{OR}}$ $\mathcal{L}_{\text{OR}} = -\log \sigma \left(\log \frac{\text{odds}(y_w x)}{\text{odds}(y_l x)} \right)$ Key Terms: ▶ $\mathcal{L}_{\text{SFT}}$: Base supervised fine-tuning loss (correct code/syntax). ▶ $\mathcal{L}_{\text{OR}}, \lambda$: Odds ratio penalty term and penalty weight. ▶ x, y_w, y_l: Prompt, winning action path, and losing action path.	Advantages: ▶ Reference-model-free: eliminates maintaining frozen π_{ref} in memory. ▶ Monolithic single-phase training avoids unstable PPO policy optimization. ▶ Directly aligns PrismSpace agent trajectories on paired feedback (<i>chosen</i> vs. <i>rejected</i>). Disadvantages: ▶ Requires high-quality pairwise preference datasets (y_w, y_l). ▶ Sensitive to λ hyperparameter balance.

Reinforcement Online Learning to Rank with Unbiased Reward Shaping

Citation	Summary	Advantages & Disadvantages
S. Zhuang, Z. Qiao, and G. Zuccon <i>Information Retrieval Journal</i> Year: 2023 Publisher: Springer Open Access	Develops an Online Learning to Rank (OLTR) framework using counterfactual reinforcement learning. Employs inverse propensity weighting (IPW) and unbiased reward shaping to correct for position, presentation, and click bias in user interaction logs during continuous ranker optimization.	Advantages: ▶ Guarantees unbiased gradient estimation from observational implicit feedback. ▶ Operates entirely online without requiring manual document relevance labels. ▶ Directly informs PrismSpace's context retrieval ranking feedback loop. Disadvantages: ▶ High propensity score variance can destabilize gradient updates. ▶ Requires large volumes of query-click logs to converge. ▶ Assumes position-based user click models that may not hold in code search.

Knowledge Learning for Securing Workflow Scheduling in **MEC** (Mobile Edge Computing)

Citation	Summary	Advantages & Disadvantages
T. M. Ghazal, A. E. Awouda, M. K. Hasan, A. H. A. Rahman, S. Islam, R. A. Saeed, H. Elshafie, A. Iqbal, and A. H. Hussein <i>Mobile Networks and Applications</i> Year: 2025 Publisher: Springer Open Access	Proposes a Knowledge-Guided Reinforcement Learning (KG-RL) scheduler for complex workflow DAGs in distributed edge-cloud environments. Integrates security constraint validation directly into the state transition model, preventing insecure task placement while optimizing makespan.	Advantages: ▶ Enforces strict security boundary constraints during automated task scheduling. ▶ Effectively handles multi-dependency Directed Acyclic Graph (DAG) workflows. ▶ Informs PrismSpace's sandboxed tool execution and dependency resolution. Disadvantages: ▶ Constraint checking introduces scheduling latency for massive DAGs. ▶ Assumes static security classification matrices across network nodes. ▶ Limited evaluation on non-deterministic tool failure scenarios.

Energy-Aware Cooperative Multi-Fitness Evolutionary Workflow Scheduling

Citation	Summary	Advantages & Disadvantages
P. Barredo and J. Puente <i>Natural Computing</i> Year: 2025 Publisher: Springer Open Access	Presents a Cooperative Multi-Fitness Evolutionary Algorithm (CMFEA) for multi-objective workflow scheduling. Optimizes trade-offs between execution time, monetary compute cost, and energy consumption using island-model cooperative coevolution across heterogeneous worker pools.	Advantages: ▶ Pareto-optimal trade-offs across makespan, financial cost, and energy. ▶ Cooperative sub-population decomposition prevents premature local convergence. ▶ Provides heuristics for PrismSpace's multi-agent model provider selection. Disadvantages: ▶ Evolutionary search has higher runtime latency than single-pass ML heuristics. ▶ Cannot adapt instantaneously to real-time API rate limit drops. ▶ Requires parameter tuning across mutation, crossover, and migration rates.

Optimized Task Scheduling Using Hybrid **DL** (Deep Learning) and Metaheuristics

Citation	Summary	Advantages & Disadvantages
A. Heirati, M. Fartash, and M. Khalily-Dermany <i>Neural Processing Letters</i> Year: 2026 Publisher: Springer Open Access	Combines Deep Learning (LSTM (Long Short-Term Memory)) workload prediction with metaheuristic optimization (Harris Hawks Optimization) for proactive workflow dispatching. Predicts task execution durations and memory bottlenecks in advance to prevent node overloading and minimize queue waiting times.	Advantages: ▶ Proactive workload forecasting prevents sudden compute node thrashing. ▶ Hybrid pipeline balances search exploration with predictive dispatching. ▶ Relevant for predicting token latency and cost before agent invocation. Disadvantages: ▶ Two-stage pipeline introduces inference overhead before scheduling decisions. ▶ LSTM prediction accuracy degrades under sudden irregular workload spikes. ▶ Complex offline model retraining requirements.

Cost-Aware Workflow Offloading in Edge-Cloud Computing via **GA** (Genetic Algorithm)

Citation	Summary	Advantages & Disadvantages
S. Abdi, M. Ashjaei, and S. Mubeen <i>The Journal of Supercomputing</i> Year: 2024 Publisher: Springer Open Access	Proposes a Genetic Algorithm-based workflow offloading framework considering heterogeneous network bandwidth, deadline constraints, and monetary costs. Implements chromosome repair operators to preserve DAG precedence validity throughout mutation and crossover.	Advantages: ▶ Guarantees valid DAG dependency structures via specialized repair operators. ▶ Explicitly accounts for network transfer latency between pipeline stages. ▶ Informs subtask payload transfer between worker agents in PrismSpace. Disadvantages: ▶ Computationally heavy for interactive real-time user query response. ▶ Assumes deterministic network transfer bandwidth in cloud environments. ▶ Does not model dynamic agent failures or retries.

Dynamic Role Discovery and Assignment in Multi-Agent Task Decomposition

Citation	Summary	Advantages & Disadvantages
Y. Xia, J. Zhu, and L. Zhu <i>Complex & Intelligent Systems</i> Year: 2023 Publisher: Springer Open Access	Proposes dynamic role discovery and assignment that decomposes complex high-level objectives into lower-level subtasks. Uses learned persona embeddings and mutual information to adaptively allocate subtasks according to situational agent capabilities, eliminating rigid static assignments.	Advantages: ▶ Enables flexible subtask allocation without hardcoded role hierarchies. ▶ Hierarchical decomposition prevents single-agent cognitive context overload. ▶ Serves as core theoretical basis for PrismSpace's dynamic agent routing. Disadvantages: ▶ Role discovery clustering introduces upfront computational overhead. ▶ Performance depends on representation vector accuracy for subtasks. ▶ Evaluated primarily on synthetic collaborative coordination benchmarks.

Decision Variable Contribution Based Adaptive Multi-Objective Scheduling

Citation	Summary	Advantages & Disadvantages
J. Li, L. Xing, W. Zhong, Z. Cai, and F. Hou <i>Complex & Intelligent Systems</i> Year: 2023 Publisher: Springer Open Access	Introduces a Decision Variable Contribution (DVC) adaptive mechanism for evolutionary multi-objective workflow scheduling. Evaluates the differential sensitivity of individual decision variables to focus evolutionary search operators on high-impact scheduling bottlenecks.	Advantages: ▶ Accelerates Pareto-front convergence by prioritizing high-variance variables. ▶ Dynamically adapts search intensity across workflow bottleneck stages. ▶ Useful for pinpointing high-latency critical path tools in agent workflows. Disadvantages: ▶ Sensitivity calculation adds per-generation matrix computation cost. ▶ Can over-concentrate on dominant objectives at the expense of diversity. ▶ Requires calibrated normalization across widely different metric units.

GHQ: Grouped Hybrid Q-Learning for Heterogeneous Multi-Agent RL

Citation	Summary	Advantages & Disadvantages
X. Yu, Y. Lin, X. Wang, S. Han, and K. Lv <i>Complex & Intelligent Systems</i> Year: 2024 Publisher: Springer Open Access	Develops Grouped Hybrid Q-learning (GHQ) for cooperative heterogeneous multi-agent systems. Clusters agents with similar capabilities into functional cohorts that share value approximations while maintaining decoupled action execution heads for role specialization.	Advantages: ▶ Scales value estimation across large heterogeneous swarms via cohort grouping. ▶ Reduces joint action space dimensionality from exponential to block-linear. ▶ Directly applicable to categorizing LLM worker agent skill specializations. Disadvantages: ▶ Cohort clustering boundaries can be rigid if agent capabilities evolve. ▶ Intra-group credit assignment remains non-trivial in dense interactions. ▶ Requires domain knowledge to establish initial grouping heuristics.

Decentralized Multi-Agent Cooperation via Adaptive Partner Modeling

Citation	Summary	Advantages & Disadvantages
C. Xu, J. Wang, X. Zhu, Y. Yue, W. Zhou, Z. Liang, and D. Wojtczak <i>Complex & Intelligent Systems</i> Year: 2024 Publisher: Springer Open Access	Tackles environmental non-stationarity in decentralized multi-agent cooperation using Adaptive Partner Modeling (APM). Instead of relying on expensive continuous inter-agent broadcast communication, agents infer partner policies and intended actions from local observation histories.	Advantages: ▶ Minimizes inter-agent communication bandwidth to near zero during execution. ▶ Stabilizes decentralized joint execution despite teammate policy shifts. ▶ Directly adopted in PrismSpace for zero-communication agent coordination. Disadvantages: ▶ Partner inference error propagates if historical observations are noisy. ▶ Inference networks introduce local computational and memory overhead. ▶ Assumes partners follow relatively stationary parameterized policy spaces.

A Responsible AI Framework: Pipeline Contextualisation

Citation	Summary	Advantages & Disadvantages
E. Vyhmeister, G. G. Castañé, P.-O. Östberg, and S. Thevenin <i>AI and Ethics</i> Year: 2023 Publisher: Springer Open Access	Formulates an operational Responsible AI Framework providing pipeline contextualisation across the full model lifecycle. Specifies auditable checkpoints, data lineage tracking, and bias mitigation protocols for autonomous AI decision-making systems.	Advantages: ▶ Holistic governance framework linking technical architecture to ethical safety. ▶ Defines concrete telemetry points for data provenance and action auditing. ▶ Guides PrismSpace's execution audit logging and human inspection UI. Disadvantages: ▶ Qualitative framework that requires custom translation into concrete code gates. ▶ Audit logging introduces storage overhead and slight execution latency. ▶ Does not provide automated formal verification of LLM reasoning traces.

Meaningful Human Control: Actionable Properties for AI Development

Citation	Summary	Advantages & Disadvantages
L. Cavalcante Siebert, M. L. Lupetti, E. Aizenberg, N. Beckers, A. Zgonnikov, H. Veluwenkamp, D. Abbink, E. Giaccardi, G.-J. Houben, C. M. Jonker, J. van den Hoven, D. Forster, and R. L. Legendijk <i>AI and Ethics</i> Year: 2023 Publisher: Springer Open Access	Formalizes the philosophical concept of Meaningful Human Control (MHC) into actionable software design properties: Tracking (system responsive to human moral reasons) and Tracing (system actions causally attributable to accountable human supervisors).	Advantages: ▶ Translates abstract governance principles into rigorous software specifications. ▶ Establishes tracking and tracing requirements for autonomous tool execution. ▶ Directly guides the design of PrismSpace's Human-in-the-Loop (HITL) approval gates. Disadvantages: ▶ High human intervention frequency can degrade automated agent throughput. ▶ Defining actionable reason boundaries requires careful domain calibration. ▶ Assumes human supervisor maintains full comprehension of complex agent plans.

CPN-HRL (Cloud-Private-Network Hierarchical Reinforcement Learning): Hierarchical **DRL** for Priority-Aware Task Scheduling

Citation	Summary	Advantages & Disadvantages
Q. Yue, L. Tian, X. Feng, Z. Yuan, S. Wei, J. Xia, B. Chen, X. Song, and Y. Yao <i>Journal of King Saud University CIS</i> Year: 2026 Publisher: Springer Open Access	Presents CPN-HRL, a Hierarchical Reinforcement Learning framework for priority-aware task scheduling in Cloud-Edge-End networks. Uses a macro-policy to partition task DAGs by priority and latency budgets, while micro-policies schedule subtasks onto optimal compute nodes.	Advantages: ▶ Hierarchical policy decomposition effectively handles complex task DAGs. ▶ Guarantees priority-aware quality of service under tight latency deadlines. ▶ Serves as primary reference for PrismSpace's DAG workflow scheduling engine. Disadvantages: ▶ Multi-tier policy architecture requires synchronous training coordination. ▶ Edge node failure requires immediate micro-policy replanning. ▶ Evaluated in simulated telecom cloud-edge topologies.

Automatic Discovery and Description of Human Planning Strategies

Citation	Summary	Advantages & Disadvantages
J. Skirzyński, Y. Jain, and F. Lieder <i>Behavior Research Methods</i> Year: 2023 Publisher: Springer Open Access	Develops an algorithmic framework for automatically discovering and formalizing planning strategies from human behavior traces. Models task planning as resource-rational search over Directed Graphical Models, producing interpretable decision trees that characterize human heuristic planning.	Advantages: ▶ Extracts interpretable, human-like heuristic planning strategies automatically. ▶ Quantifies trade-offs between cognitive search effort and decision quality. ▶ Informs PrismSpace's hierarchical prompt decomposition and agent planning. Disadvantages: ▶ Search space for strategy discovery grows exponentially with problem depth. ▶ Human behavioral experiments conducted in stylized cognitive testbeds. ▶ Static decision trees struggle to capture fluid LLM reasoning trajectories.

Data-Intensive Workflow Scheduling Based on **DRL** in Multi-Clouds

Citation	Summary	Advantages & Disadvantages
S. Zhang, Z. Zhao, C. Liu, and S. Qin <i>Journal of Cloud Computing</i> Year: 2023 Publisher: Springer Open Access	Proposes a Deep Q-Network (DQN) scheduler optimized for data-intensive workflows across multi-cloud environments. Explicitly models dataset placement and cross-datacenter transfer latency to minimize both workflow completion time and data egress billing.	Advantages: ▶ Optimizes data locality to prevent massive cross-cloud egress overhead. ▶ Adaptive reward function balances monetary cost and makespan constraints. ▶ Directly relevant to caching and embedding retrieval across distributed agents. Disadvantages: ▶ Assumes deterministic file sizes and network bandwidth availability. ▶ Q-network value overestimation requires double-DQN correction. ▶ State representation size scales with the total number of cloud datacenters.

RL-Based Task Scheduling for Sustainable Federated Cloud Computing

Citation	Summary	Advantages & Disadvantages
Z. Wang, S. Chen, L. Bai, J. Gao, J. Tao, R. R. Bond, and M. D. Mulvenna <i>Journal of Cloud Computing</i> Year: 2023 Publisher: Springer Open Access	Develops an Actor-Critic RL scheduling framework for federated cloud datacenters that incorporates renewable energy availability into scheduling policies. Dynamically routes compute tasks to datacenters with green energy surpluses without violating user SLA deadlines.	Advantages: ▶ Incorporates environmental sustainability and carbon metrics into scheduling. ▶ Actor-Critic policy gradient handles continuous action spaces effectively. ▶ Provides conceptual foundation for multi-attribute provider routing. Disadvantages: ▶ Requires real-time telemetry from green energy generation sources. ▶ Trade-off between carbon footprint and network transmission latency. ▶ Sensitive to reward scaling factors between SLA penalties and green rewards.

Hybrid Dolphin Swarm Sparrow Search Multi-Objective Workflow Scheduling

Citation	Summary	Advantages & Disadvantages
S. Subashree, M. Rajakumaran, G. Pushpa, and S. Senthilkumar <i>Journal of Cloud Computing</i> Year: 2025 Publisher: Springer Open Access	Proposes a hybrid bio-inspired metaheuristic combining Dolphin Swarm Optimization (DSA) and Sparrow Search Algorithm (SSA) for multi-objective cloud workflow scheduling. Optimizes execution makespan, virtual machine utilization, and monetary execution expenditure.	Advantages: ▶ Combines global exploration of DSA with local exploitation of SSA. ▶ Demonstrates superior Pareto hypervolume compared to standard GA and PSO (Particle Swarm Optimization). ▶ Provides strong comparative baselines for workflow optimization algorithms. Disadvantages: ▶ Metaheuristic iteration runtime is too slow for sub-second reactive routing. ▶ Multiple tuning parameters (predation rate, vocalization index) required. ▶ Lacks theoretical convergence rate bounds for stochastic optimization.

Research Gap

What the surveyed methods already do

MARL works address **CTDE**, role grouping, and ad-hoc coordination. **ORPO** aligns **LLM** preferences without reference models, while **HIAT** leverages human interventions as auxiliary **RL** signals. **CPN-HRL** and **DAG** schedulers optimize cloud task allocation, and **MHC** formalizes tracking and tracing properties.

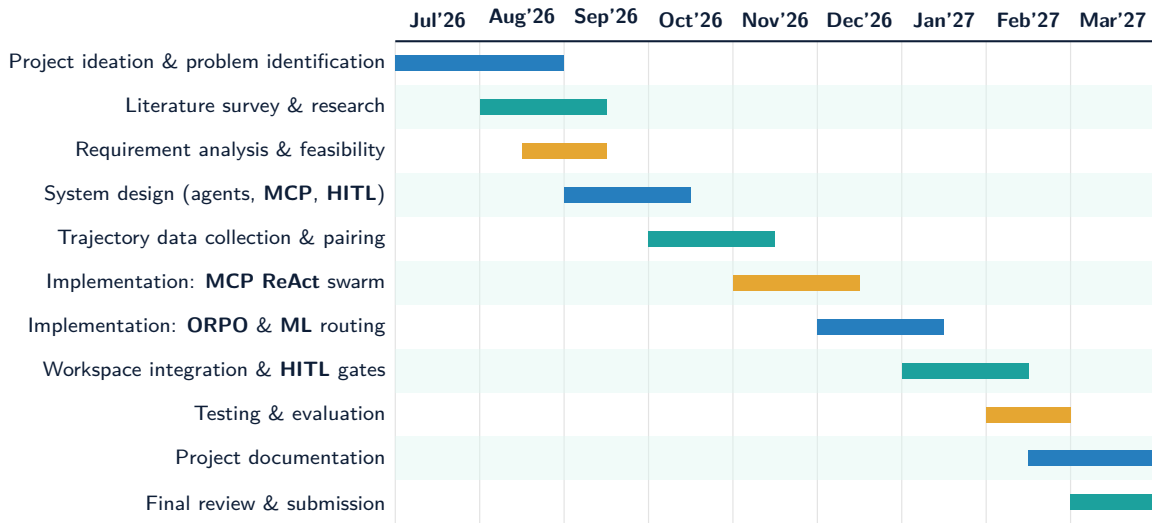
Limitations observed in those papers

Coordination benchmarks focus on gaming (**SMAC**, Overcooked) rather than filesystem/**API** tooling. **ORPO** lacks grounding in **MCP ReAct** tool traces, while **HIAT** targets continuous robotic control instead of code execution. Cloud schedulers assume static **DAGs** or slow metaheuristics ill-suited for real-time **LLM** routing.

Why PrismSpace is required

No reviewed system jointly grounds tool observations, aligns policies on agent trajectories, dynamically routes heterogeneous **LLMs**, and enforces human-in-the-loop safety gates within a unified developer workspace.

Gantt Chart



Timeline (Jul 2026 – Mar 2027)

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Thank You

Questions & Discussion

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